SHEET INDEX

40 kW AC-DC (lower) - 1 of 3

AC-SENSING - 9 sections

AUX-POWER - 5 sections

CONTROL - 4 sections

DC-LINK - 4 sections

INPUT-EMI - 4 sections

VIENNA-PFC - 3 sections

rev D.4 - 29 sections - 320 components

NET NAMING

internal junctions are named for what they join

U<ref>_<PIN> node at that IC pin e.g. UIVOA_VINP

R<stem>_M midpoint of a series pair e.g. RBALTA_M

R<stem>_<nm> tap between R<stem>n/m e.g. RV1D_01

all others are explicit design nets

MAGNETICS CONSTRUCTION

identity per docs/magnetics.md (turns, gap, litz, acceptance lines live there)

D1 LA0-LC0: 5x 0077908A7 KoolMu (CATALOG AL37), N=26 +/-1 lot trim -> L0 116uH, >=61uH @104A pk (E51)

E68 EMI: CMC1 + 12x X2 4.7uF in 3 star stages + CMC2 + RC damper; D6 DM chokes deleted

Cr at TANK sections

40 kW ACDC board 1of2 - Vienna PFC (1x hot cells)

DC-Modules 40 kW - board AC-DC (lower), sheet 1 of 3

2026-09-06 · rev D.4